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To the Editor of [journal name — to be chosen by the authors],

Submission: "Choosing a Fast Stand-In for a Slow Simulation: a Selection Tool and a 44-Problem Study"

The practical problem. Engineers who rely on a slow simulation can afford only a limited number of runs, and then have to decide which fast stand-in model to train and whether more runs would be worth the cost. Published comparisons of Gaussian-process, reduced-basis and neural-network stand-ins disagree, are often run under unequal tuning, and do not give a procedure a practitioner can run on their own data.

What the paper offers. (1) An open tool, SurroSelect, that chooses a stand-in by cross-validation, reports which models are tied, checks whether the POD basis limits accuracy, and forecasts the error for more runs. (2) A pre-specified evaluation of the tool (criteria committed to version control before the evaluation runs) on eight problems solved by an independent finite-element library, against a tuned neural network, and on two real measured datasets. (3) A study of 44 problems describing where each family wins, with controls for the reduced basis, the regressor and the width of the hyper-parameter search.

The evidence, in numbers. Of the seven pre-specified hypotheses for the tool and the benchmark finding, five were met, two were not met and zero could not be evaluated (details in the paper). The tool did not match the benchmark's own selection among methods that include networks, and the benchmark finding that a reduced-basis model wins at large budgets held on only three of the eight problems. Selection regret against the oracle had median 1.00 (95% interval 1.00–1.02); the tuned network was better than the tool by more than 25% in four of 16 cells; the learning-curve forecast was within a factor two of the actual error in 77% of 48 cases. Limitations are listed in the paper: the problems are simulated, the real-data check uses scalar noisy outputs, the networks were tuned with a limited budget, and the forecast is a rough extrapolation.

Why this journal. [To be written by the authors after choosing the journal and checking its scope, article types, indexing and author guidelines; the project record does not contain a verified match for the authors' constraints.]

Statements. The manuscript is original and is not under review elsewhere [authors to confirm]. All data are generated by the code in the project repository or are public, openly licensed measurements (CC BY 4.0); the tool is released under the MIT licence; a permanent archive with a DOI [to be provided]. The study design, code, analyses and drafting used an AI system (Claude) under the authors' direction, as disclosed in the acknowledgments; the internal reviews were AI-agent reviews, not human peer review; all authors must approve the submission.

Sincerely,
[Author names, affiliations, corresponding-author e-mail — to be supplied by the authors]